

HIKONE, Japan

WMO: 477610

Lat: 35.2758N

Lon: 136.2436E

Elev: 93

StdP: 100.22

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-1.3	-0.4	-6.8	2.1	2.8	-5.5	2.4	2.9	11.0	2.5	10.0	3.4	3.3	150	0.403

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	7.2	33.4	25.6	32.4	25.3	31.2	24.9	26.4	31.6	25.9	30.9	25.4	30.0	3.5	310

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
25.0	20.3	29.6	24.4	19.6	29.0	23.9	19.0	28.4	83.0	31.7	80.6	30.9	78.4	30.1	28.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9.2	8.2	7.2		-3.2	35.3	1.2	1.0	-4.1	36.0	-4.8	36.6	-5.4	37.1	-6.3	37.8
				-4.2	27.5	1.1	0.5	-5.0	27.9	-5.7	28.2	-6.3	28.5	-7.1	28.9
				DB											
				WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	15.3	3.8	4.3	7.3	12.7	18.0	22.1	26.6	28.0	23.9	17.9	11.7	6.5
	DBStd	8.74	2.03	2.38	2.83	3.16	2.51	2.29	2.35	1.89	2.74	2.74	2.80	2.62
	HDD10.0	583	192	162	92	11	0	0	0	0	0	0	14	113
	HDD18.3	2000	450	394	341	170	37	1	0	0	1	40	199	368
	CDD10.0	2520	0	1	9	92	248	363	515	559	417	247	65	4
	CDD18.3	895	0	0	0	1	27	114	257	301	168	28	0	0
	CDH23.3	7456	0	0	0	6	97	502	2381	3289	1117	64	0	0
	CDH26.7	2501	0	0	0	1	8	84	831	1300	272	4	0	0
Wind	WSAvg	2.8	3.4	3.4	3.2	2.7	2.5	2.3	2.3	2.5	2.7	2.8	2.8	3.2
Precipitation	PrecAvg	1821	163	124	124	113	151	177	220	139	190	134	123	165
	PrecMax	2272	270	176	185	197	406	323	411	427	405	276	217	318
	PrecMin	1295	62	45	60	38	61	77	58	30	65	44	55	83
	PrecStd	218	56	32	35	44	81	62	101	88	90	54	42	55
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.9	13.3	18.6	24.0	27.7	30.9	34.4	34.8	32.5	27.0	20.1	15.3
		MCWB	8.0	9.8	12.4	15.8	18.7	23.5	25.8	25.8	24.6	21.7	15.8	12.4
	2%	DB	9.5	10.9	15.7	21.5	25.8	28.9	33.1	33.7	30.9	25.0	18.6	13.3
		MCWB	6.7	7.8	10.9	14.5	17.7	22.2	25.4	25.7	24.3	20.1	14.9	10.6
	5%	DB	8.4	9.5	13.8	19.7	24.2	27.3	32.0	32.8	29.5	23.7	17.5	12.0
		MCWB	5.9	6.9	10.0	14.1	17.2	21.4	25.2	25.5	23.6	19.0	14.3	9.3
	10%	DB	7.4	8.4	12.0	18.0	22.7	26.1	30.7	31.9	28.2	22.4	16.4	10.8
		MCWB	5.0	6.0	8.8	13.4	16.7	21.0	24.8	25.2	23.0	18.0	13.5	8.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.0	10.8	13.7	18.3	21.1	24.9	27.1	27.3	26.1	22.7	17.4	13.4
		MCDB	10.1	12.3	16.4	20.6	24.5	29.3	32.5	32.7	30.6	25.5	18.8	14.6
	2%	WB	7.3	8.7	12.1	16.5	19.5	23.9	26.2	26.7	25.1	21.1	16.1	11.2
		MCDB	8.8	10.1	14.8	19.9	23.2	27.1	31.3	32.0	29.1	24.0	17.7	12.8
	5%	WB	6.3	7.5	10.7	15.0	18.5	22.9	25.7	26.2	24.4	20.0	15.0	9.7
		MCDB	8.0	9.0	13.1	18.4	22.2	26.0	30.5	31.3	28.2	22.6	16.9	11.5
	10%	WB	5.2	6.2	9.3	13.8	17.6	21.9	25.1	25.7	23.7	18.9	13.9	8.5
		MCDB	7.1	8.2	11.5	16.8	21.2	24.9	29.7	30.5	27.3	21.6	15.9	10.5
Mean Daily Temperature Range	5% DB	MDBR	5.5	5.9	7.3	8.5	8.3	6.8	6.6	7.2	6.7	7.1	6.9	6.0
		MCDBR	6.8	7.8	10.6	11.3	11.1	8.8	8.3	8.2	7.7	8.0	7.5	7.3
		MCWBR	5.0	5.5	6.9	6.0	5.3	4.1	3.5	3.3	3.5	4.2	5.0	5.4
	5% WB	MCDBR	6.4	7.0	8.8	8.9	8.5	7.0	7.6	7.5	6.9	6.4	6.2	6.5
		MCWBR	5.2	5.6	6.6	5.5	4.8	3.6	3.5	3.3	3.5	4.0	5.0	5.5
Clear-Sky Solar Irradiance	taub	0.409	0.443	0.504	0.536	0.542	0.566	0.561	0.534	0.485	0.448	0.418	0.406	
	taud	2.090	2.018	1.896	1.871	1.919	1.931	1.999	2.066	2.171	2.188	2.172	2.119	
	Ebn at Noon	756	777	764	764	765	745	745	757	776	768	745	731	
	Edh at Noon	125	150	185	199	192	190	176	162	139	126	115	115	
All-Sky Solar Radiation	RadAvg	2.08	2.78	3.79	4.68	5.17	4.73	4.83	4.98	3.95	3.21	2.38	1.87	
	RadStd	0.15	0.24	0.29	0.43	0.55	0.42	0.70	0.49	0.35	0.30	0.20	0.16	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (38 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page